Lab 1 - Internet Protocols

Part 1: HTTP

Task 1:

Full Connection:

Protocol	Length	Info
TCP	78	49413 → 80 [SYN, ECE, CWR] Seq=0 Win=65535 Len=0 MSS=1460 WS=64 TSval=3038597837 TSecr=0 SACK_PERM
TCP	78	49414 → 80 [SYN, ECE, CWR] Seq=0 Win=65535 Len=0 MSS=1460 WS=64 TSval=1803358789 TSecr=0 SACK_PERM
TCP	78	[TCP Retransmission] 49414 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=64 TSval=1803359790 TSecr=0 SACK_PERM
TCP	78	[TCP Retransmission] 49413 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=64 TSval=3038598838 TSecr=0 SACK_PERM
TCP	78	[TCP Retransmission] 49414 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=64 TSval=1803360790 TSecr=0 SACK_PERM
TCP	78	[TCP Retransmission] 49413 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=64 TSval=3038599838 TSecr=0 SACK_PERM
TCP	78	[TCP Retransmission] 49414 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=64 TSval=1803361792 TSecr=0 SACK_PERM
TCP	78	[TCP Retransmission] 49413 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=64 TSval=3038600840 TSecr=0 SACK_PERM
TCP	74	80 → 49413 [SYN, ACK] Seq=0 Ack=1 Win=26847 Len=0 MSS=1412 SACK_PERM TSval=4243082735 TSecr=3038600840 WS=128
TCP	66	49413 → 80 [ACK] Seq=1 Ack=1 Win=131648 Len=0 TSval=3038601143 TSecr=4243082735
HTTP	578	GET /online HTTP/1.1
TCP	66	80 → 49413 [ACK] Seq=1 Ack=513 Win=27136 Len=0 TSval=4243083023 TSecr=3038601143
HTTP	601	HTTP/1.1 301 Moved Permanently (text/html)
TCP	66	49413 → 80 [ACK] Seq=513 Ack=536 Win=131136 Len=0 TSval=3038601542 TSecr=4243083141
HTTP	579	GET /online/ HTTP/1.1
TCP	66	80 → 49413 [ACK] Seq=536 Ack=1026 Win=28160 Len=0 TSval=4243083428 TSecr=3038601550
TCP	1354	80 → 49413 [ACK] Seq=536 Ack=1026 Win=28160 Len=1288 TSval=4243083428 TSecr=3038601550 [TCP PDU reassembled in…
HTTP	297	HTTP/1.1 200 OK (text/html)
TCP	66	49413 → 80 [ACK] Seq=1026 Ack=2055 Win=129664 Len=0 TSval=3038601777 TSecr=4243083428
HTTP	510	GET /favicon.ico HTTP/1.1
TCP	78	[TCP Retransmission] 49414 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=64 TSval=1803362793 TSecr=0 SACK_PERM
HTTP	482	HTTP/1.1 200 OK (PNG)
TCP	66	49413 → 80 [ACK] Seq=1470 Ack=2471 Win=130688 Len=0 TSval=3038602050 TSecr=4243083700
TCP	78	[TCP Retransmission] 49414 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=64 TSval=1803363794 TSecr=0 SACK_PERM
TCP	74	80 → 49414 [SYN, ACK] Seq=0 Ack=1 Win=26847 Len=0 MSS=1412 SACK_PERM TSval=4243084719 TSecr=1803363794 WS=128
TCP	66	49414 → 80 [ACK] Seq=1 Ack=1 Win=131648 Len=0 TSval=1803364033 TSecr=4243084719
TCP	66	49413 → 80 [FIN, ACK] Seq=1470 Ack=2471 Win=131072 Len=0 TSval=3038603872 TSecr=4243083700
TCP	66	49414 → 80 [FIN, ACK] Seq=1 Ack=1 Win=131648 Len=0 TSval=1803364824 TSecr=4243084719
TCP	66	80 → 49413 [FIN, ACK] Seq=2471 Ack=1471 Win=29312 Len=0 TSval=4243085749 TSecr=3038603872
TCP	66	49413 → 80 [ACK] Seq=1471 Ack=2472 Win=131072 Len=0 TSval=3038604101 TSecr=4243085749
TCP	66	80 → 49414 [FIN, ACK] Seq=1 Ack=2 Win=26112 Len=0 TSval=4243085750 TSecr=1803364824
TCP	66	49414 → 80 [ACK] Seq=2 Ack=2 Win=131648 Len=0 TSval=1803365055 TSecr=4243085750

Task 2:

Filtering HTTP packets:

http

No.	Time	Protocol	Length	Info
645	8.541905	HTTP	601	GET /online HTTP/1.1
655	8.779771	HTTP	627	HTTP/1.1 301 Moved Permanently (text/html)
659	8.784926	HTTP	602	GET /online/ HTTP/1.1
688	9.020086	HTTP	225	HTTP/1.1 200 OK (text/html)
693	9.064979	HTTP	536	GET /favicon.ico HTTP/1.1
731	9.355714	HTTP	502	HTTP/1.1 200 OK (PNG)

HTTP 200 respone message:

```
Hypertext Transfer Protocol
  > HTTP/1.1 200 OK\r\n
    Date: Mon, 02 Feb 2026 13:37:39 GMT\r\n
    Server: Apache/2.4.62 ()\r\n
    Last-Modified: Wed, 29 Jun 2022 00:26:43 GMT\r\n
    ETag: "7c-5e28b351ffe27"\r\n
    Accept-Ranges: bytes\r\n
  > Content-Length: 124\r\n
    Keep-Alive: timeout=5, max=98\r\n
    Connection: Keep-Alive\r\n
    Content-Type: image/vnd.microsoft.icon\r\n
    \r\n
    [Request in frame: 693]
    [Time since request: 290.735000 milliseconds]
    [Request URI: /favicon.ico]
    [Full request URI: http://lushbeautifulinnermagic.neverssl.com/favicon.ico]
    File Data: 124 bytes
```

Part 2: TCP/IP

Task 1:

Filtering TCP packets:

tcp

No.	Time	Protocol	Length	Info
568	6.520122	TCP	78	[TCP Retransmission] 49593 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=64 TSval=570340594 TSecr=0 SACK_PERM
570	6.535864	TCP	86	443 → 49591 [ACK] Seq=4514 Ack=2662 Win=69632 Len=0 TSval=2906972970 TSecr=3332731380
591	7.220056	TCP	98	[TCP Retransmission] 49590 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1412 WS=64 TSval=478240945 TSecr=0 SACK_PERM
592	7.220110	TCP	98	[TCP Retransmission] 49589 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1412 WS=64 TSval=3610747452 TSecr=0 SACK_PERM
606	7.520992	TCP	78	[TCP Retransmission] 49594 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=64 TSval=1245092626 TSecr=0 SACK_PERM
607	7.521060	TCP	78	[TCP Retransmission] 49593 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=64 TSval=570341595 TSecr=0 SACK_PERM
624	8.182635	TCP	86	49591 → 443 [FIN, ACK] Seq=2662 Ack=4514 Win=131072 Len=0 TSval=3332733289 TSecr=2906972970
629	8.220975	TCP	98	[TCP Retransmission] 49590 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1412 WS=64 TSval=478241946 TSecr=0 SACK_PERM
630	8.221027	TCP	98	[TCP Retransmission] 49589 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1412 WS=64 TSval=3610748453 TSecr=0 SACK_PERM
638	8.447922	TCP	86	443 → 49591 [ACK] Seq=4514 Ack=2663 Win=69632 Len=0 TSval=2906974882 TSecr=3332733289
641	8.521157	TCP	78	[TCP Retransmission] 49594 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=64 TSval=1245093626 TSecr=0 SACK_PERM
642	8.521224	TCP	78	[TCP Retransmission] 49593 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=64 TSval=570342595 TSecr=0 SACK_PERM
643	8.541313	TCP	94	80 → 49590 [SYN, ACK] Seq=0 Ack=1 Win=26847 Len=0 MSS=1392 SACK_PERM TSval=3701852439 TSecr=478241946 WS=128
644	8.541477	TCP	86	49590 → 80 [ACK] Seq=1 Ack=1 Win=131136 Len=0 TSval=478242266 TSecr=3701852439
645	8.541905	HTTP	601	GET /online HTTP/1.1
654	8.779769	TCP	86	80 → 49590 [ACK] Seq=1 Ack=516 Win=28032 Len=0 TSval=3701852678 TSecr=478242267
655	8.779771	HTTP	627	HTTP/1.1 301 Moved Permanently (text/html)
656	8.779944	TCP	86	49590 → 80 [ACK] Seq=516 Ack=542 Win=130624 Len=0 TSval=478242505 TSecr=3701852678
659	8.784926	HTTP	602	GET /online/ HTTP/1.1
687	9.020084	TCP	1466	80 → 49590 [ACK] Seq=542 Ack=1032 Win=29056 Len=1380 TSval=3701852918 TSecr=478242510 [TCP PDU reassembled in 688]
688	9.020086	HTTP	225	HTTP/1.1 200 OK (text/html)
689	9.020238	TCP	86	49590 → 80 [ACK] Seq=1032 Ack=2061 Win=129600 Len=0 TSval=478242745 TSecr=3701852918
693	9.064979	HTTP	536	GET /favicon.ico HTTP/1.1
718	9.222369	TCP	98	[TCP Retransmission] 49589 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1412 WS=64 TSval=3610749454 TSecr=0 SACK_PERM
731	9.355714	HTTP	502	HTTP/1.1 200 OK (PNG)
732	9.355892	TCP	86	49590 → 80 [ACK] Seq=1482 Ack=2477 Win=130688 Len=0 TSval=478243081 TSecr=3701853204
745	9.522475	TCP	78	[TCP Retransmission] 49593 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=64 TSval=570343596 TSecr=0 SACK_PERM
773	10.223601	TCP	98	[TCP Retransmission] 49589 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1412 WS=64 TSval=3610750456 TSecr=0 SACK_PERM
787	10.522724	TCP	78	[TCP Retransmission] 49593 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=64 TSval=570344597 TSecr=0 SACK_PERM

Task 2:

The TCP three-way handshake:

TCP	78	[TCP Retransmission] 49413 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=64 TSval=3038600840 TSecr=0 SACK_PERM
TCP	74	80 → 49413 [SYN, ACK] Seq=0 Ack=1 Win=26847 Len=0 MSS=1412 SACK_PERM TSval=4243082735 TSecr=3038600840 WS=128
TCP	66	49413 → 80 [ACK] Seq=1 Ack=1 Win=131648 Len=0 TSval=3038601143 TSecr=4243082735

TCP Stream:

```
GET /online HTTP/1.1
Host: lushbeautifulinnermagic.neverssl.com
Connection: keep-alive
Upgrade-Insecure-Requests: 1
User-Agent: Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/144.0.0.0 Safari/537.36
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.7
Referer: http://neverssl.com/
Accept-Encoding: gzip, deflate
Accept-Language: ar-AE,ar;q=0.9,en-US;q=0.8,en;q=0.7


HTTP/1.1 301 Moved Permanently
Date: Mon, 02 Feb 2026 13:37:39 GMT
Server: Apache/2.4.62 ()
Location: http://lushbeautifulinnermagic.neverssl.com/online/
Content-Length: 259
Keep-Alive: timeout=5, max=100
Connection: Keep-Alive
Content-Type: text/html; charset=iso-8859-1

<!DOCTYPE HTML PUBLIC "-//IETF//DTD HTML 2.0//EN">
<html><head>
<title>301 Moved Permanently</title>
</head><body>
<h1>Moved Permanently</h1>
<p>The document has moved <a href="http://lushbeautifulinnermagic.neverssl.com/online/">here</a>.</p>
</body></html>

GET /online/ HTTP/1.1
Host: lushbeautifulinnermagic.neverssl.com
Connection: keep-alive
Upgrade-Insecure-Requests: 1
User-Agent: Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/144.0.0.0 Safari/537.36
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.7
Referer: http://neverssl.com/
Accept-Encoding: gzip, deflate
Accept-Language: ar-AE,ar;q=0.9,en-US;q=0.8,en;q=0.7


HTTP/1.1 200 OK
Date: Mon, 02 Feb 2026 13:37:39 GMT
Server: Apache/2.4.62 ()
Last-Modified: Wed, 29 Jun 2022 00:23:22 GMT
ETag: "8be-5e28b29291e10-gzip"
Accept-Ranges: bytes
Vary: Accept-Encoding
Content-Encoding: gzip
```

Packet 659. 3 client pkts, 4 server pkts, 5 turns. Click to select.

Entire conversation (3957 bytes) | Show as ASCII | No delta times | Stream 14

Find: | Case sensitive | Find Next

Help | Filter Out This Stream | Print | Save as... | Back | Close

Observed data packets:

```
> Transmission Control Protocol, Src Port: 49413, Dst Port: 80, Seq: 1, Ack: 1, Len: 512
v Hypertext Transfer Protocol
  v GET /online HTTP/1.1\r\n
      Request Method: GET
      Request URI: /online
      Request Version: HTTP/1.1
    Host: oldfreshclearpathway.neverssl.com\r\n
    Connection: keep-alive\r\n
    Upgrade-Insecure-Requests: 1\r\n
    User-Agent: Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.36 (KHTML, lik
    Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/
    Referer: http://neverssl.com/\r\n
    Accept-Encoding: gzip, deflate\r\n
    Accept-Language: ar-AE,ar;q=0.9,en-US;q=0.8,en;q=0.7\r\n
    \r\n
    [Response in frame: 857]
    [Full request URI: http://oldfreshclearpathway.neverssl.com/online]

0040  4d ef 47 45 54 20 2f 6f  6e 6c 69 6e 65 20 48 54   M·GET /o nline HT
0050  54 50 2f 31 2e 31 0d 0a  48 6f 73 74 3a 20 6f 6c   TP/1.1·· Host: ol
0060  64 66 72 65 73 68 63 6c  65 61 72 70 61 74 68 77   dfreshcl earpathw
0070  61 79 2e 6e 65 76 65 72  73 73 6c 2e 63 6f 6d 0d   ay.never ssl.com·
0080  0a 43 6f 6e 6e 65 63 74  69 6f 6e 3a 20 6b 65 65   ·Connect ion: kee
0090  70 2d 61 6c 69 76 65 0d  0a 55 70 67 72 61 64 65   p-alive· ·Upgrade
00a0  2d 49 6e 73 65 63 75 72  65 2d 52 65 71 75 65 73   -Insecur e-Reques
00b0  74 73 3a 20 31 0d 0a 55  73 65 72 2d 41 67 65 6e   ts: 1··U ser-Agen
00c0  74 3a 20 4d 6f 7a 69 6c  6c 61 2f 35 2e 30 20 28   t: Mozil la/5.0 (
00d0  4d 61 63 69 6e 74 6f 73  68 3b 20 49 6e 74 65 6c   Macintos h; Intel
00e0  20 4d 61 63 20 4f 53 20  58 20 31 30 5f 31 35 5f    Mac OS  X 10_15_
00f0  37 29 20 41 70 70 6c 65  57 65 62 4b 69 74 2f 35   7) Apple WebKit/5
0100  33 37 2e 33 36 20 28 4b  48 54 4d 4c 2c 20 6c 69   37.36 (K HTML, li
0110  6b 65 20 47 65 63 6b 6f  29 20 43 68 72 6f 6d 65   ke Gecko ) Chrome
0120  2f 31 34 34 2e 30 2e 30  2e 30 20 53 61 66 61 72   /144.0.0 .0 Safar
0130  69 2f 35 33 37 2e 33 36  0d 0a 41 63 63 65 70 74   i/537.36 ··Accept
0140  3a 20 74 65 78 74 2f 68  74 6d 6c 2c 61 70 70 6c   : text/h tml,appl
0150  69 63 61 74 69 6f 6e 2f  78 68 74 6d 6c 2b 78 6d   ication/ xhtml+xm
0160  6c 2c 61 70 70 6c 69 63  61 74 69 6f 6e 2f 78 6d   l,applic ation/xm
0170  6c 3b 71 3d 30 2e 39 2c  69 6d 61 67 65 2f 61 76   l;q=0.9, image/av
0180  69 66 2c 69 6d 61 67 65  2f 77 65 62 70 2c 69 6d   if,image /webp,im
0190  61 67 65 2f 61 70 6e 67  2c 2a 2f 2a 3b 71 3d 30   age/apng ,*/*;q=0
01a0  2e 38 2c 61 70 70 6c 69  63 61 74 69 6f 6e 2f 73   .8,appli cation/s
01b0  69 67 6e 65 64 2d 65 78  63 68 61 6e 67 65 3b 76   igned-ex change;v
01c0  3d 62 33 3b 71 3d 30 2e  37 0d 0a 52 65 66 65 72   =b3;q=0. 7··Refer
01d0  65 72 3a 20 68 74 74 70  3a 2f 2f 6e 65 76 65 72   er: http ://never
01e0  73 73 6c 2e 63 6f 6d 2f  0d 0a 41 63 63 65 70 74   ssl.com/ ··Accept
01f0  2d 45 6e 63 6f 64 69 6e  67 3a 20 67 7a 69 70 2c   -Encodin g: gzip,
0200  20 64 65 66 6c 61 74 65  0d 0a 41 63 63 65 70 74    deflate ··Accept
0210  2d 4c 61 6e 67 75 61 67  65 3a 20 61 72 2d 41 45   -Languag e: ar-AE
0220  2c 61 72 3b 71 3d 30 2e  39 2c 65 6e 2d 55 53 3b   ,ar;q=0. 9,en-US;
0230  71 3d 30 2e 38 2c 65 6e  3b 71 3d 30 2e 37 0d 0a   q=0.8,en ;q=0.7··
0240  0d 0a                                              ··
```

Connection Termination:

```
TCP    66  49413 → 80  [FIN, ACK] Seq=1470 Ack=2471 Win=131072 Len=0 TSval=3038603872 TSecr=4243083700
TCP    66  49414 → 80  [FIN, ACK] Seq=1 Ack=1 Win=131648 Len=0 TSval=1803364824 TSecr=4243084719
TCP    66  80 → 49413  [FIN, ACK] Seq=2471 Ack=1471 Win=29312 Len=0 TSval=4243085749 TSecr=3038603872
TCP    66  49413 → 80  [ACK] Seq=1471 Ack=2472 Win=131072 Len=0 TSval=3038604101 TSecr=4243085749
TCP    66  80 → 49414  [FIN, ACK] Seq=1 Ack=2 Win=26112 Len=0 TSval=4243085750 TSecr=1803364824
TCP    66  49414 → 80  [ACK] Seq=2 Ack=2 Win=131648 Len=0 TSval=1803365055 TSecr=4243085750
```

The connection closes using:

1. **FIN-ACK**: Client initiates close
2. **ACK**: Server acknowledges
3. **FIN-ACK**: Server also closes

4. **ACK**: Client acknowledges

Part 3: UDP

Task 1:

I used `nslookup google.com` in terminal because DNS lookup uses UDP

Full connection:

No.	Time	Protocol	Length	Info
1	0.000000	SSDP	212	M-SEARCH * HTTP/1.1
39	7.378175	UDP	86	57621 → 57621 Len=44
44	9.056349	DNS	90	Standard query 0x75e7 A google.com
45	9.077484	DNS	106	Standard query response 0x75e7 A google.com A 142.251.209.110
49	9.156629	UDP	1193	53475 → 443 Len=1131
51	9.232775	UDP	86	443 → 53475 Len=24
53	9.464247	UDP	493	443 → 53475 Len=431
54	9.464722	UDP	105	53475 → 443 Len=43
56	9.467570	DNS	80	Standard query 0xdbc8 HTTPS beacons.gcp.gvt2.com
57	9.467801	DNS	80	Standard query 0xd564 AAAA beacons.gcp.gvt2.com
58	9.467936	DNS	80	Standard query 0x3abf A beacons.gcp.gvt2.com
59	9.505107	DNS	167	Standard query response 0xdbc8 HTTPS beacons.gcp.gvt2.com CNAME beacons-handoff.gcp.gvt2.com SOA ns1.google.com
60	9.539497	DNS	138	Standard query response 0xd564 AAAA beacons.gcp.gvt2.com CNAME beacons-handoff.gcp.gvt2.com AAAA 2607:f8b0:4002:c10:
61	9.539499	DNS	126	Standard query response 0x3abf A beacons.gcp.gvt2.com CNAME beacons-handoff.gcp.gvt2.com A 34.128.8.163

Task 2:

Filtered UDP Packets:

udp

No.	Time	Protocol	Length	Info
1	0.000000	SSDP	212	M-SEARCH * HTTP/1.1
39	7.378175	UDP	86	57621 → 57621 Len=44
44	9.056349	DNS	90	Standard query 0x75e7 A google.com
45	9.077484	DNS	106	Standard query response 0x75e7 A google.com A 142.251.209.110
49	9.156629	UDP	1193	53475 → 443 Len=1131
51	9.232775	UDP	86	443 → 53475 Len=24
53	9.464247	UDP	493	443 → 53475 Len=431
54	9.464722	UDP	105	53475 → 443 Len=43
56	9.467570	DNS	80	Standard query 0xdbc8 HTTPS beacons.gcp.gvt2.com
57	9.467801	DNS	80	Standard query 0xd564 AAAA beacons.gcp.gvt2.com
58	9.467936	DNS	80	Standard query 0x3abf A beacons.gcp.gvt2.com
59	9.505107	DNS	167	Standard query response 0xdbc8 HTTPS beacons.gcp.gvt2.com CNAME beacons-handoff.gcp.gvt2.com SOA ns1.google.com
60	9.539497	DNS	138	Standard query response 0xd564 AAAA beacons.gcp.gvt2.com CNAME beacons-handoff.gcp.gvt2.com AAAA 2607:f8b0:4002:c10::5e
61	9.539499	DNS	126	Standard query response 0x3abf A beacons.gcp.gvt2.com CNAME beacons-handoff.gcp.gvt2.com A 34.128.8.163
62	9.541405	QUIC	1292	Initial, DCID=e25acffc34b15269, PKN: 1, PING, CRYPTO, CRYPTO
63	9.541447	QUIC	1292	Initial, DCID=e25acffc34b15269, PKN: 2, PING, CRYPTO
64	9.541711	QUIC	141	0-RTT, DCID=e25acffc34b15269
65	9.717925	QUIC	102	Initial, SCID=e25acffc34b15269, PKN: 1, ACK
66	9.717928	QUIC	1292	Initial, SCID=e25acffc34b15269, PKN: 2, ACK, PADDING

UDP Header:

```
User Datagram Protocol, Src Port: 57621, Dst Port: 57621
    Source Port: 57621
    Destination Port: 57621
    Length: 52
    Checksum: 0x8108 [unverified]
    [Checksum Status: Unverified]
    [Stream index: 1]
    [Stream Packet Number: 1]
  > [Timestamps]
    UDP payload (44 bytes)
```

Part 4: Comparing TCP and UDP

Task 1: Fill in the following table and provide reasons.

Feature	TCP/UDP	Reasons
Reliability and Connection Establishment	TCP	TCP is connection-oriented and establishes a reliable connection through the three-way handshake (SYN, SYN-ACK, ACK). It guarantees delivery of data by using acknowledgments and retransmissions for lost packets. UDP is connectionless and does not establish a connection or guarantee delivery.
Data Integrity and Ordering	TCP	TCP ensures data integrity using checksums and guarantees that data arrives in the correct order using sequence numbers. If packets arrive out of order, TCP reorders them. UDP provides only basic checksum error detection and does not guarantee packet ordering - packets may arrive out of order or not at all.

Task 2: Identify the use Cases and Performance of TCP and UDP.

	TCP	UDP
Use cases	• Web browsing (HTTP/HTTPS) • Email (SMTP, IMAP, POP3) • File transfers (FTP, SFTP) • Remote access (SSH, Telnet) • Database queries • Any application requiring reliable data delivery	• Live streaming (video/audio) • Online gaming • VoIP (Voice over IP) • DNS queries • IoT sensor data • Video conferencing • Any application where speed > reliability
Performance	**Slower but reliable:** • Higher overhead due to connection establishment • Acknowledgments and retransmissions add latency • Flow control and congestion control prevent network overload • Guaranteed delivery increases processing time • Best for applications where accuracy matters more than speed	**Faster but unreliable:** • Minimal overhead (no handshake, no acknowledgments) • Lower latency - data sent immediately • No retransmissions means less delay • No flow/congestion control • Smaller header size (8 bytes vs 20+ bytes) • Best for real-time applications where some data loss is acceptable